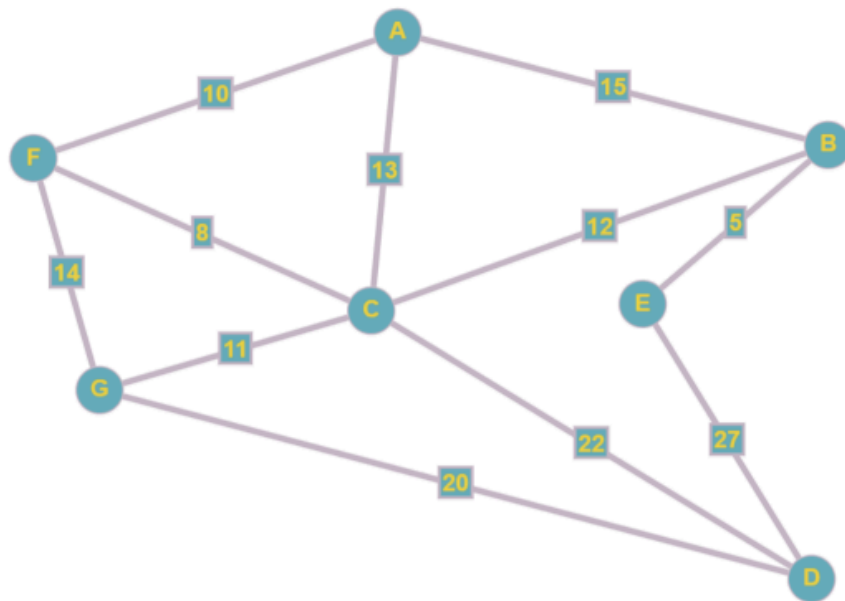




Nodes	H(n)
A	20
B	18
C	15
D	0
E	12
F	11
H	11
G	12

Q1. Apply A* search to find the shortest path. (Tree Search)

Q2. Check whether node E is admissible.

Q3

Two admissible heuristics $h_1(n)$ and $h_2(n)$ are given. For all nodes, $h_2(n) \geq h_1(n)$, and both heuristics are also consistent.

If A* uses $h_2(n)$ instead of $h_1(n)$, how will it affect:

- (a) the completeness of A* search, and
- (b) the number of node expansions?

Justify both answers.

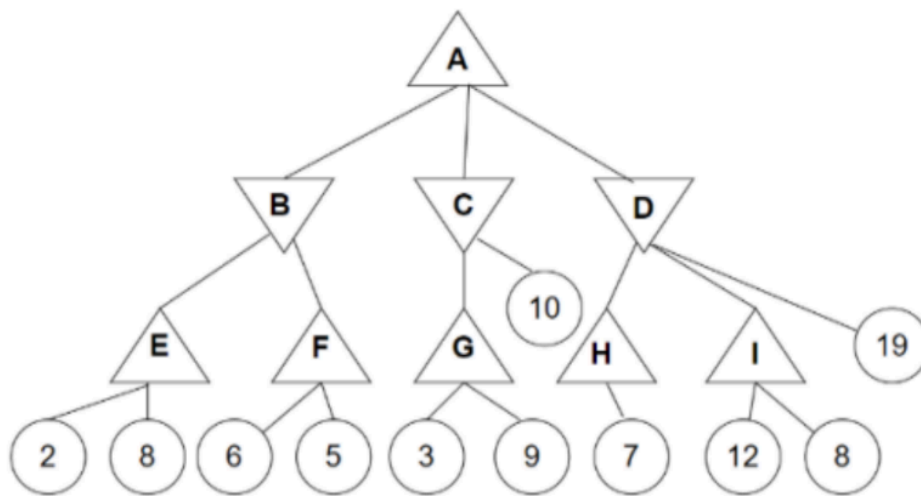
[3 marks]

1. A shopping mall faces customer congestion issues at the escalators during peak hours. Each floor has multiple escalators going up and down, but not all are equally used. To improve flow and reduce bottlenecks, the mall management wants to optimize **escalator direction (up/down), operation time, and speed settings** across various floors. The aim is to **minimize waiting time for shoppers and avoid overcrowding**, while keeping energy consumption low.

Each escalator can operate in different configurations:

- **Direction:** Up (1) or Down (0)
 - **Operation Time:** [Low, Medium, High] (encoded as 0, 1, 2)
 - **Speed:** [Slow, Normal, Fast] (encoded as 0, 1, 2)
- a. Formulate the problem using a Genetic Algorithm, showing 4 chromosomes [2]
 - b. Suggest a possible fitness function [1]
 - c. What is the purpose of Selection, Crossover, and Mutation? Show each of them on your chromosomes [2]

2. Apply Alpha Beta Pruning on the tree above.



3. A robot vacuum is navigating through a large, cluttered apartment to find its charging dock before the battery runs out. The robot has limited sensors—it can only detect obstacles and floor type in its immediate surroundings. The apartment has different zones like carpet, tiles, and raised floor areas. The robot does not know the full layout and must decide which way to move step by step.

- a. Explain how the robot might use the hill-climbing algorithm to reach the charging dock.
- b. State two issues with the hill climbing algorithm.
- c. State how simulated annealing works in this case.